### APPH 6202. Clinical Gait Analysis. 3 Credit Hours.

Analysis of normal and pathological human locomotion. Study of theory and instrumentation for measurement of temporal and spatial kinematics and kinetics, electromyography, and plantar pressure.

### APPH 6203. Biomechanics and Kinesiology in Prosthetics and Orthotics. 2 Credit Hours.

Mechanics of human movement applied to the study of artifical limbs and braces. Emphasis on neuromuscular control, Newtonian mechanics, kinematics and kinetics.

### APPH 6209. Clinical Pathology. 2 Credit Hours.

Systems level overview of human pathology with emphasis on the effect of disease on human movement and neuromechanical function.

### APPH 6211. Systems Physiology I: Cellular Mechanisms of Plasticity. 3 Credit Hours.

The course will focus on adaptations of skeletal, muscular, and neural systems at the cellular level.

### APPH 6212. Systems Physiology II: Physiology of Neuromotor Tissues. 3 Credit Hours.

The course will focus on function and adaptations of skeletal, muscular, and neural systems. Interactions among the various systems and their plasticity will be emphasized.

### APPH 6213. Systems Physiology III: Integrated Systems and Adaptation. 3 Credit Hours.

The course will focus on integrative mechanism impacting motor system performance. Interactions among the various systems and their plasticity will be emphasized.

### APPH 6214. Laboratory Rotations in Prosthetics and Orthotics. 2 Credit Hours.

This course will provide the opportunity for students in individual laboratories to support their graduate training in prosthetics and orthotics.

### APPH 6215. Studies in Responsible Conduct of Research in Prosthetics and Orthotics. 3 Credit Hours.

This course will cover areas related to research ethics, the responsible use of animial and human models and collaborative research issues in prosthetics and orthotics.